DO NOW — RECALL

Paper 2 examiner-style feedback

Name: _____ Class: _____ Date: _____

1. Why must recursive calls progress?

2. What does a midpoint calculate?

3. What happens when low>high?

4. C# [2,4,6] with a[1]=a[0]+a[2]. Final array?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

To reach the base case.

2. Model answer

The middle index of remaining interval.

3. Model answer

No candidates remain; not found.

4. Model answer

[2,8,6].

By the end, you can…

- Identify whether the error lies in problem reading, algorithm state or pseudocode.
- Explain every bound update, recursive return or data-structure operation.
- Retest the method with a different input to demonstrate transfer.

Vocabulary — know these terms

mark scheme — Criteria for awarding marks.

termination — Algorithm eventually stops.

auxiliary space — Additional storage beyond the input.

NEW LEARNING

Read, annotate and explain

Understand

Identify whether the error lies in problem reading, algorithm state or pseudocode.

Apply

Explain every bound update, recursive return or data-structure operation.

Evaluate

Retest the method with a different input to demonstrate transfer.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Repair a SumTo recursion missing a base case.

Explain the result

The n==0 base case stops recursion and returns zero.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

An algorithm's recursion has no base case. Explain fault and give two repairs. [6]

Question 2

Trace binary search for 9 in [1,3,5,7,9], recording every low, high and midpoint. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Correct: 'merge sort has O(1) auxiliary space because the temp array is short-lived'. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Getting a correct output once guarantees an algorithm is bug-free.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.